QUANTUM FIELD THEORY IN DE SITTER SPACE

Foundations, Challenges, and Implications for Physics

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Abstract

Quantum Field Theory (QFT) in curved spacetime provides an essential bridge between quantum mechanics and general relativity, with de Sitter (dS) space serving as the canonical arena. De Sitter space approximates both the inflationary epoch of the early universe and the late-time accelerated expansion driven by dark energy, making a rigorous quantum treatment of fields in this background physically necessary. Yet this setting presents deep conceptual challenges absent in flat spacetime: the absence of a global time direction produces vacuum ambiguity, observer-dependent particle creation, and thermal effects—the Gibbons–Hawking radiation at temperature $T = H/2\pi$—alongside infrared divergences that threaten perturbative stability.

This paper examines two complementary frameworks that address these difficulties. The first, *Direct-Sum Quantum Field Theory* (DQFT), reconstructs the quantum description by elevating the causal structure of de Sitter space to a structural principle in Hilbert space, recovering unitarity and thermality simultaneously. The second, the *Ambient Space Formalism*, embeds de Sitter space as a hyperboloid in five-dimensional flat spacetime, enabling representation-theoretic construction of fields with manifest de Sitter invariance. By comparing these approaches, we illuminate how modern formulations resolve longstanding tensions between covariance, unitarity, and observer dependence, and we discuss implications for inflationary cosmology, quantum gravity, and open problems in the field.

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1 Introduction

The problem of formulating quantum field theory on a curved spacetime background has occupied theoretical physics for over half a century. While the discovery of Hawking radiation [1] and the Unruh effect [2] established that geometry and thermodynamics are inextricably linked in quantum theory, the case of de Sitter space poses challenges that are both more fundamental and more practically urgent. Observational evidence strongly indicates that the universe has twice undergone extended phases of de Sitter-like expansion: once during primordial inflation, whose quantum fluctuations seeded the large-scale structure of the cosmos [3, 4], and again in the present era of dark energy-driven acceleration [5]. A quantum field theory adequate to this background is therefore not merely of academic interest.

The central obstruction is geometric. In Minkowski space, global time-translation symmetry selects a preferred vacuum state and a canonical definition of particles. In de Sitter space, no globally defined timelike Killing vector exists; the cosmological horizon confines every observer to a finite causal patch, and the notion of a particle becomes observer-dependent. Gibbons and Hawking [5] showed that a static observer detects thermal radiation at temperature

$$T_{\mathrm{dS}} = \frac{H}{2\pi},$$

where $H = \sqrt{\Lambda/3}$ is the Hubble parameter. This result is analogous to Hawking radiation, but arises not from gravitational collapse but from cosmic expansion itself.

Subsequent work revealed additional pathologies. Allen and Mottola [6, 7] identified a one-parameter family of de Sitter-invariant vacua (the α-vacua), undermining uniqueness. Ford and Parker [8] found infrared divergences for massless minimally coupled fields, and Tsamis and Woodard [9] showed that graviton interactions can produce secular back-reaction. These issues prompted a search for frameworks that go beyond standard mode-decomposition methods.

Two such frameworks form the focus of this paper. Section 2 reviews the geometry of de Sitter space and the causal structure that underlies thermality. Section 3 summarises the standard apparatus of QFT in curved spacetime and its failure modes in de Sitter space. Sections 4 and 5 develop the two main approaches: Direct-Sum Quantum Field Theory (DQFT) and the Ambient Space Formalism, respectively. Section 6 discusses their physical implications and broader connections to cosmology and quantum gravity, and Section 7 concludes.

2 De Sitter Spacetime

2.1 Geometric Definition and the Embedding Picture

De Sitter space is the unique maximally symmetric Lorentzian manifold with positive constant curvature. It arises as the vacuum solution of the Einstein field equations

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 0$$

with a positive cosmological constant $\Lambda > 0$. The three maximally symmetric vacuum solutions—Minkowski ($\Lambda = 0$), de Sitter ($\Lambda > 0$), and anti-de Sitter ($\Lambda < 0$)—are compared schematically in Fig. 1.

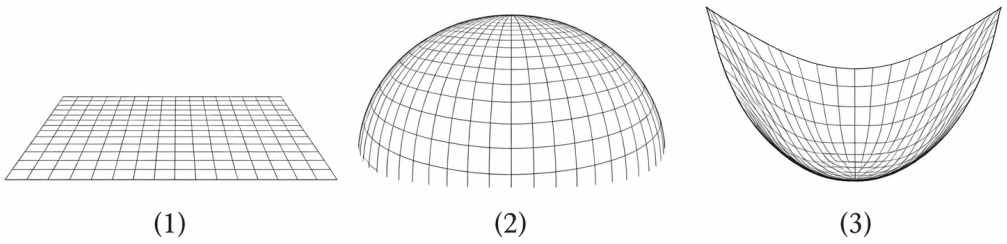


Figure 1: The three maximally symmetric vacuum spacetimes distinguished by the sign of Λ. Minkowski space ($\Lambda = 0$) is flat; de Sitter space ($\Lambda > 0$) has positive curvature and undergoes exponential expansion; anti-de Sitter space ($\Lambda < 0$) has negative curvature and a timelike boundary.

The most transparent description of de Sitter space is through its embedding as a four-dimensional hyperboloid in five-dimensional Minkowski space $\mathbb{R}^{1,4}$ [10]. With ambient coordinates $X^A = (X^0, X^1, X^2, X^3, X^4)$ and flat metric $\eta_{AB} = \mathrm{diag}(-1, +1, +1, +1, +1)$, de Sitter space dS_4 is the locus

$$\eta_{AB} X^A X^B = -X_0^2 + X_1^2 + X_2^2 + X_3^2 + X_4^2 = H^{-2},$$

where $H^2 = \Lambda/3$. The induced metric on this hyperboloid is the de Sitter metric with Hubble parameter H. The embedding makes the symmetry group $SO(1,4)$ manifest: any linear isometry of $\mathbb{R}^{1,4}$ that preserves the hyperboloid is a symmetry of de Sitter space. De Sitter space is therefore the homogeneous space $SO(1,4)/SO(1,3)$.

2.2 Coordinate Systems

Different classes of observers in de Sitter space are most naturally described by different coordinate charts. We introduce the two most physically relevant.

Flat (inflationary) coordinates. For comoving observers, de Sitter space takes the FLRW form

$$ds^2 = -dt^2 + e^{2Ht}\left(dx^2 + dy^2 + dz^2\right),$$

covering only one half of the global hyperboloid ($-\infty < t < \infty$, $\mathbf{x} \in \mathbb{R}^3$). These coordinates are standard in inflationary cosmology, where $a(t) = e^{Ht}$ is the scale factor and H the constant expansion rate. Introducing conformal time $\eta = -e^{-Ht}/H \in (-\infty, 0)$, the metric becomes

$$ds^2 = \frac{1}{H^2\eta^2}\left(-d\eta^2 + d\mathbf{x}^2\right),$$

manifestly conformally flat. These coordinates are the natural setting for computing inflationary perturbation spectra [4, 3].

Static coordinates. For a geodesic observer at rest at the origin, de Sitter space takes the static form

$$ds^2 = -(1 - H^2r^2)\,dt^2 + \frac{dr^2}{1 - H^2r^2} + r^2\,d\Omega^2,$$

covering the static patch $r < H^{-1}$. The coefficient $1 - H^2r^2$ vanishes at $r = H^{-1}$, which is the cosmological horizon. These coordinates are the de Sitter analogue of Schwarzschild coordinates and are especially useful for analyzing horizon thermality.

2.3 Causal Structure and the Cosmological Horizon

A defining feature of de Sitter space is the presence of a cosmological event horizon for every timelike observer. Because the scale factor $a(t) = e^{Ht}$ grows exponentially, any comoving point beyond the Hubble radius H^{-1} is causally disconnected from the observer: light emitted there can never reach the observer's worldline. Unlike a black hole horizon, the de Sitter horizon is observer-dependent—each observer carries their own horizon centered on their worldline.

The global causal structure, most clearly seen via a Penrose diagram, shows that the full de Sitter

manifold has both a future and a past cosmological horizon for every observer. No single observer has access to the full hyperboloid. The static patch accessible to an observer at $r = 0$ is the intersection of their past and future light cones, bounded by the horizon at $r = H^{-1}$.

2.4 Gibbons–Hawking Thermality

The most striking quantum consequence of the de Sitter horizon was established by Gibbons and Hawking in 1977 [5]. Even if the global quantum state of a field is the de Sitter-invariant (Bunch–Davies) vacuum, a static observer confined to their causal patch detects a thermal bath of radiation at temperature

$$T_{\rm dS} = \frac{H}{2\pi}. \tag{1}$$

The mechanism is identical in structure to Hawking radiation: quantum modes of the field are entangled across the horizon. Since the observer cannot access degrees of freedom beyond $r = H^{-1}$, tracing over those inaccessible modes yields a mixed state with a thermal density matrix

$$\rho_+ = \mathrm{Tr}_-(|\Psi\rangle\langle\Psi|) \propto e^{-H_{\rm static}/T_{\rm dS}},$$

where $H_{\rm static}$ generates time translations in the static patch. The de Sitter horizon also carries an entropy

$$S_{\rm dS} = \frac{A}{4G}, \qquad A = 4\pi H^{-2},$$

suggesting that only a finite number of degrees of freedom are physically accessible within any causal patch.

3 Quantum Field Theory in Curved Spacetime: Background and Challenges

3.1 Fields on Curved Backgrounds

Quantum field theory in curved spacetime (QFTCS) treats the spacetime metric as a fixed classical background on which quantum fields propagate [11, 12, 13]. The starting point is a classical field: a function $\phi : \mathcal{M} \to \mathbb{R}$ on a Lorentzian manifold $(\mathcal{M}, g_{\mu\nu})$ obeying a covariant wave equation. For a real

scalar field, the action in curved spacetime is

$$S[\phi] = -\frac{1}{2}\int d^4x\,\sqrt{-g}\,\left[g^{\mu\nu}\nabla_\mu\phi\,\nabla_\nu\phi + (m^2+\xi R)\phi^2\right],$$

where R is the Ricci scalar and ξ is a curvature coupling. Variation yields the Klein–Gordon equation on curved spacetime:

$$\left(\Box_g - m^2 - \xi R\right)\phi = 0, \qquad \Box_g = \frac{1}{\sqrt{-g}}\partial_\mu\left(\sqrt{-g}\,g^{\mu\nu}\partial_\nu\right).$$

Quantization proceeds by expanding the field in a complete set of modes $\{u_k(x)\}$,

$$\phi(x) = \sum_k \left(a_k\,u_k(x) + a_k^\dagger\,u_k^*(x)\right),$$

and promoting the expansion coefficients to operators satisfying canonical commutation relations $[a_k, a_{k'}^\dagger] = \delta_{kk'}$.

3.2 The Vacuum Ambiguity

In Minkowski space, the vacuum is the unique Poincaré-invariant state, and positive-frequency modes are unambiguously defined by $u_k \propto e^{i\mathbf{k}\cdot\mathbf{x}-i\omega_k t}$ with $\omega_k > 0$. In a general curved spacetime with no preferred timelike Killing vector, there is no canonical decomposition into positive and negative frequencies. Different mode choices $\{u_k\}$ and $\{\tilde{u}_k\}$ are related by Bogoliubov transformations [11],

$$\tilde{u}_k = \sum_{k'}\left(\alpha_{kk'}\,u_{k'} + \beta_{kk'}\,u_{k'}^*\right),$$

and the corresponding Fock vacua $|0\rangle$ and $|\tilde{0}\rangle$ differ whenever the Bogoliubov coefficients $\beta_{kk'} \neq 0$. An observer using $\tilde{u}_k$ sees $|0\rangle$ as containing particles:

$$\langle 0|\tilde{N}_k|0\rangle = \sum_{k'}|\beta_{kk'}|^2 \neq 0.$$

In de Sitter space, this multiplicity is especially severe: Allen and Mottola showed that there is a one-parameter family of de Sitter-invariant vacua, the *α-vacua* [6, 7], each yielding different physical predictions for particle creation rates and correlation functions. The Bunch–Davies vacuum corresponds to $\alpha = -\infty$ and is singled out by the requirement that modes reduce to Minkowski positive-frequency modes in the

short-distance limit.

3.3 Infrared Divergences

A further difficulty is the accumulation of infrared modes in de Sitter space. For massless minimally coupled scalar fields ($m = 0$, $\xi = 0$), the two-point function grows logarithmically with the conformal time separation [8, 14]:

$$\langle 0|\phi(x)\phi(x')|0\rangle \sim -\frac{H^2}{4\pi^2}\ln\left[-H^2(x-x')^2\right] + \cdots$$

This infrared growth reflects a de Sitter-space analogue of the zero-mode problem. At loop level, such divergences can produce secular effects—terms growing as powers of the expansion factor $a(t)$—that potentially destabilize the de Sitter background [9]. Whether these instabilities are genuine physical effects or artifacts of perturbation theory remains under debate [15, 16].

3.4 Observer Dependence and the Loss of Unitarity

Perhaps the deepest issue is the apparent tension between observer dependence and unitarity. When a static observer in de Sitter space describes the field using their local Hamiltonian, the global de Sitter-invariant state appears as a thermal density matrix. Density matrices of this form conventionally signal mixed, non-pure states—apparently in conflict with unitary time evolution. Understanding how thermal behavior and unitarity can coexist in de Sitter space motivates the two frameworks to which we now turn.

4 Direct-Sum Quantum Field Theory (DQFT)

4.1 Motivation: The Problem of Time in de Sitter Space

Standard quantum field theory presupposes a global time coordinate that defines a Hamiltonian and an associated notion of energy. In de Sitter space, as emphasized in the previous section, no such global time exists. Different observers, related by de Sitter isometries, adopt inequivalent time slicings, leading to inequivalent Hilbert spaces and incompatible particle definitions.

A natural response—pursued in the DQFT framework of Kumar and Marto [17]—is to abandon the attempt to impose a single time on all of de Sitter space and instead allow the quantum description to accommodate both time orientations as co-equal, mutually exclusive structures. The key insight is that de Sitter space has two inequivalent time orientations that are not related by any continuous symmetry of

the spacetime; they are related only by the discrete operation of time reversal. Quantum theory should therefore incorporate both on equal footing.

4.2 Direct-Sum Hilbert Space

The DQFT framework begins by replacing the standard single Hilbert space $\mathcal{H}$ with a direct sum

$$\mathcal{H}_{\rm DQFT} = \mathcal{H}^{(+)} \oplus \mathcal{H}^{(-)}, \tag{2}$$

where $\mathcal{H}^{(+)}$ corresponds to the forward time orientation (the sector accessible to a given static observer) and $\mathcal{H}^{(-)}$ corresponds to the complementary sector beyond the cosmological horizon. This is not a tensor product: states in $\mathcal{H}^{(+)}$ and $\mathcal{H}^{(-)}$ are mutually exclusive; no physical process can create a superposition across the two sectors. The direct sum encodes the causal disconnection intrinsic to de Sitter space.

Observables are represented by operators of the block-diagonal form

$$\hat{O} = \begin{pmatrix} \hat{O}^{(+)} & 0 \\ 0 & \hat{O}^{(-)} \end{pmatrix},$$

acting on $\mathcal{H}^{(+)}$ and $\mathcal{H}^{(-)}$ independently. This block structure encodes the superselection rule imposed by the causal horizon: measurements performed by an observer in one sector cannot influence the other sector.

4.3 Field Quantization and Fock Structure

The scalar field ϕ satisfying the Klein–Gordon equation $(\Box_g - m^2)\phi = 0$ is quantized independently in each sector. We introduce mode functions $\{u_k^{(+)}\}$ adapted to the forward time orientation and $\{u_k^{(-)}\}$ adapted to the reversed orientation, each forming a complete orthonormal set with respect to the Klein–Gordon inner product. The field operators in each sector are

$$\hat{\phi}_{\pm}(x) = \sum_k \left(a_k^{(\pm)}\, u_k^{(\pm)}(x) + a_k^{(\pm)\dagger}\, u_k^{(\pm)*}(x) \right), \tag{3}$$

acting on $\mathcal{H}^{(\pm)}$ respectively. The full field operator on $\mathcal{H}_{\rm DQFT}$ is

$$\hat{\Phi}(x) = \begin{pmatrix} \hat{\phi}_+(x) & 0 \\ 0 & \hat{\phi}_-(x) \end{pmatrix}.$$

Within each sector, the operators satisfy the usual canonical commutation relations:

$$[a_k^{(\pm)},\, a_{k'}^{(\pm)\dagger}] = \delta_{kk'}, \qquad [a_k^{(\pm)},\, a_{k'}^{(\pm)}] = 0.$$

Operators from different sectors commute. Each sector independently reproduces the standard Fock space structure of flat-space QFT.

The vacuum states are defined sector-by-sector:

$$a_k^{(+)}\,|0\rangle_+ = 0 \quad \forall\, k, \qquad a_k^{(-)}\,|0\rangle_- = 0 \quad \forall\, k, \tag{4}$$

and the global DQFT vacuum is

$$|0\rangle_{\rm DQFT} = |0\rangle_+ \oplus |0\rangle_-\,. \tag{5}$$

This is not a superposition of two vacua but a single global state encoding the existence of two mutually exclusive quantum descriptions. By choosing the mode functions $u_k^{(\pm)}$ to reduce to Minkowski positive-frequency modes at short distances, the DQFT vacuum reproduces the Bunch–Davies vacuum within each sector.

4.4 Unitarity and the Origin of Thermality

The central achievement of the DQFT framework is a clean derivation of Gibbons–Hawking thermality without invoking mixed states at the fundamental level.

The global vacuum state can be written in terms of modes adapted to the static patch. A Bogoliubov transformation relating global and static patch modes produces a highly entangled state correlating excitations across the horizon:

$$|\Psi\rangle = \prod_\omega \mathcal{N}_\omega \sum_{n=0}^{\infty} e^{-\beta\omega n/2}\, \left|n_\omega^{(+)}\right\rangle \otimes \left|n_\omega^{(-)}\right\rangle, \tag{6}$$

where $\beta = 2\pi/H$. This is a pure state: $\rho_{\rm tot} = |\Psi\rangle\langle\Psi|$ satisfies $\rho_{\rm tot}^2 = \rho_{\rm tot}$.

Nevertheless, an observer confined to $\mathcal{H}^{(+)}$ cannot access or measure the $\mathcal{H}^{(-)}$ degrees of freedom. Their physical description is given by the reduced density matrix obtained by partial trace:

$$\rho_+ = \mathrm{Tr}_{(-)}(|\Psi\rangle\langle\Psi|) = \prod_\omega \mathcal{N}_\omega^2 \sum_{n=0}^{\infty} e^{-\beta\omega n} \left|n_\omega^{(+)}\right\rangle\left\langle n_\omega^{(+)}\right|. \tag{7}$$

This is precisely the thermal density matrix with inverse temperature $\beta = 2\pi/H$, reproducing the Gibbons–Hawking temperature (1). Crucially, thermality here is not a fundamental property of the global state—it is a consequence of causal restriction. The global description is unitary; thermality is an emergent phenomenon arising from the observer's inability to access the complementary sector.

4.5 CPT Symmetry and Discrete Symmetries

The two-sector structure has a natural interpretation in terms of discrete symmetries. The time-reversal operator $\mathcal{T}$ acts by exchanging the sectors: $\mathcal{T} : \mathcal{H}^{(+)} \leftrightarrow \mathcal{H}^{(-)}$. Combined with the CP transformation (which leaves each sector invariant for a neutral scalar), the full CPT symmetry of the DQFT framework exchanges forward and backward time sectors while reversing all internal quantum numbers. This is consistent with the CPT theorem [18], which DQFT preserves at a structural level. The existence of the two sectors therefore reflects the physical reality that de Sitter space has no globally preferred time orientation, and a complete quantum description must accommodate both.

5 The Ambient Space Formalism

5.1 Symmetry as a Starting Principle

The DQFT approach addressed the problem of observer dependence from an information-theoretic perspective. The ambient space formalism [19, 20] takes a complementary route, using the maximal symmetry of de Sitter space as the primary organising principle.

The guiding observation is that intrinsic coordinate descriptions of de Sitter space are necessarily observer-dependent: different coordinate patches are needed for different observers, and the symmetry group $SO(1,4)$ is not manifest in any single chart. The ambient space formalism resolves this by working in the five-dimensional flat embedding space throughout. Because $\mathbb{R}^{1,4}$ is a linear space, the action of $SO(1,4)$ is simply matrix multiplication. All fields, differential operators, and mode equations can be written in ambient coordinates, making de Sitter covariance manifest.

5.2 Geometric Setup

We recall the embedding: de Sitter space dS_4 is the hyperboloid

$$\eta_{AB} X^A X^B = H^{-2}, \qquad A, B = 0, 1, 2, 3, 4.$$

in $\mathbb{R}^{1,4}$. The tangent space at any point $X \in dS_4$ consists of all ambient vectors V^A satisfying $X_A V^A = 0$ (orthogonal to the position vector with respect to η).

Physical fields on dS_4 are represented by ambient tensor fields subject to two conditions:

1. **Homogeneity.** A field of homogeneity degree σ satisfies

$$X^A \partial_A \, \Phi(X) = \sigma \, \Phi(X).$$

   This ensures that Φ is effectively defined on the hyperboloid rather than depending on the unphysical radial direction.

2. **Tangentiality** (for tensor fields). A vector (or higher-rank) ambient field $K^A(X)$ must satisfy $X_A K^A = 0$ so that it has no component in the normal direction, i.e. no "fifth-dimensional" degrees of freedom.

Together, these constraints guarantee that the ambient field encodes exactly the physical degrees of freedom of a field on dS_4, without overcounting.

5.3 Casimir Operator and Field Equations

The generators of the de Sitter group $SO(1,4)$ act on ambient fields as

$$L_{AB} = i \left(X_A \partial_B - X_B \partial_A \right).$$

These operators satisfy the $\mathfrak{so}(1,4)$ commutation relations $[L_{AB}, L_{CD}] = i(\eta_{AC} L_{BD} - \eta_{AD} L_{BC} - \eta_{BC} L_{AD} + \eta_{BD} L_{AC})$ identically, making the symmetry algebra transparent.

The quadratic Casimir operator is

$$\mathcal{Q} = \frac{1}{2} L^{AB} L_{AB} = -H^{-2} \Box_{dS},$$

where $\Box_{dS}$ is the Laplace–Beltrami operator on dS_4. For a scalar field of mass m and curvature coupling ξ, the covariant Klein–Gordon equation

$$\left(\Box_{dS} - m^2 - \xi R\right)\Phi = 0, \qquad R = 12H^2,$$

translates in ambient language to

$$\left(\Box_{\mathbb{R}^{1,4}} - \sigma(\sigma+4)H^2\right)\Phi = 0, \tag{8}$$

where the homogeneity degree σ is related to the mass by

$$\sigma(\sigma+4) = -\frac{m^2 + \xi R}{H^2}.$$

Equation (8) is a wave equation on flat five-dimensional space, easily analysed using standard plane-wave techniques.

Remark 5.1. The mass parameter m^2 in (8) can be positive (principal series), zero (massless field), or satisfy $0 < m^2 < 2H^2$ (complementary series). In the last case, the field equation is perfectly regular in the ambient sense despite what might superficially appear as tachyonic behavior. The ambient formalism therefore makes the representation-theoretic content of de Sitter fields transparent [19, 20].

5.4 Mode Expansion and Two-Point Functions

In the ambient formalism, the Hilbert space of a free scalar field is constructed from solutions to (8) that satisfy the homogeneity and tangentiality conditions. The two-point (Wightman) function of the Bunch–Davies vacuum takes a particularly elegant form in ambient coordinates:

$$W(X, X') = \langle 0|\Phi(X)\Phi(X')|0\rangle = \mathcal{W}\big(Z(X, X')\big), \tag{9}$$

where

$$Z(X, X') = H^2\, \eta_{AB} X^A X'^B$$

is the geodesic invariant (the de Sitter analogue of the Minkowski invariant $(x - x')^2$). The Wightman function depends only on Z, reflecting the full $SO(1,4)$ invariance of the state. Explicit expressions for

$\mathcal{W}(Z)$ in terms of hypergeometric functions ${}_2F_1$ can be found in [19].

The factored form (9) has the important consequence that infrared behavior is governed entirely by the analytic properties of $\mathcal{W}(Z)$ in the limit $Z \to 1$ (coincident points) and $Z \to -\infty$ (antipodal limit). In particular, the zero-mode problem for massless minimally coupled scalars corresponds to a pole in $\mathcal{W}(Z)$ at the special mass value $m = 0$, $\xi = 0$, and the ambient formalism makes this pole structure transparent as a failure of the homogeneity condition [14].

5.5 Vector Fields, Gauge Fields, and Higher Spin

A major strength of the ambient space formalism is its natural extension to higher-spin fields. For a spin-1 gauge field (photon or graviton in linearized gravity), one introduces an ambient vector field $K^A(X)$ subject to tangentiality $X_A K^A = 0$, homogeneity with degree σ, and a gauge condition

$$\partial_A K^A(X) = 0 \quad \text{(ambient Lorenz gauge)}.$$

The physical degrees of freedom are those K^A orthogonal to pure-gauge configurations $K^A = \partial^A f$ for homogeneous scalars f. The transverse-traceless components for a spin-2 graviton are similarly isolated by additional constraints. In both cases, the de Sitter-covariant wave equation is the direct ambient analogue of the flat-space massless wave equation, and the gauge algebra is encoded in simple linear algebra on ambient fields [20].

This clean separation of physical from unphysical degrees of freedom—difficult to achieve in intrinsic coordinates due to gauge-fixing complications—is one of the main technical advantages of the ambient formalism over coordinate-based approaches.

6 Physical Implications and Comparison of Frameworks

6.1 Unitarity and Thermality: Two Perspectives

The DQFT and ambient space formalisms approach the problem of de Sitter QFT from opposite ends. DQFT prioritises unitarity and the quantum structure of state space, deriving thermality as a consequence of partial tracing over causally inaccessible degrees of freedom. The ambient formalism prioritises symmetry and covariance, constructing fields as objects transforming under irreducible representations of $SO(1,4)$ and deriving their dynamics from the Casimir equation.

These perspectives are complementary. DQFT explains *why* a static observer perceives a thermal state: because the global pure state $|\Psi\rangle$ of (6) is entangled across the horizon, and an observer confined to one sector can only access a subsystem. The ambient formalism explains *which* fields are consistent with de Sitter symmetry and provides explicit closed-form expressions for their correlation functions without fixing coordinates. Together they provide a more complete understanding than either approach alone.

6.2 Inflationary Perturbations and the CMB

The most direct observational consequence of QFT in de Sitter space concerns the primordial power spectrum. Quantum fluctuations of a nearly massless scalar field during inflation are stretched by the accelerating expansion to superhorizon scales, freezing out as classical perturbations that later source the temperature anisotropies in the cosmic microwave background (Fig. 2) [3, 4].

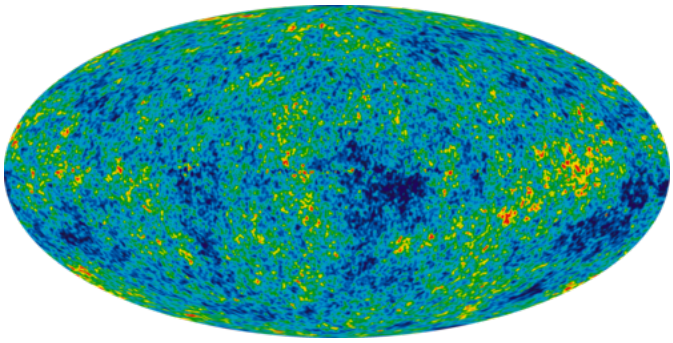


Figure 2: All-sky map of the cosmic microwave background (CMB) temperature anisotropies. The near-uniform temperature provides strong evidence for the cosmological principle on large scales, while the small fluctuations encode the primordial quantum perturbations generated during inflation. [21]

In flat de Sitter coordinates, the scalar power spectrum for a massless field is scale-invariant:

$$\mathcal{P}_\phi(k) = \frac{H^2}{4\pi^2},$$

a prediction confirmed at the level of a few percent by CMB observations. The ambient formalism makes the $SO(1,4)$ invariance of this result explicit, while the DQFT framework clarifies why the result is the same for all comoving observers despite the absence of a global time.

6.3 Infrared Stability and the de Sitter Conjecture

The infrared divergences discussed in Section 3 have prompted the question of whether de Sitter space can be a stable background for quantum gravity. Marolf and Morrison [15] showed that for interacting massive scalar fields, infrared effects do not destabilise the de Sitter vacuum at all orders in perturbation theory. However, for gravitons and massless minimally coupled scalars—where the zero-mode problem is most severe—the question remains open [9, 16].

The ambient formalism is informative here: the zero-mode pathology corresponds precisely to an indecomposable (non-semisimple) representation of $SO(1,4)$ at the massless minimally coupled value, where the standard field equation fails to select a unique irreducible module [14, 20]. This representation-theoretic diagnosis separates genuinely physical infrared effects from artifacts of mode-decomposition choices.

6.4 Observer Dependence and Quantum Gravity

Both frameworks highlight a feature that any deeper theory of quantum gravity must confront: physical observables in de Sitter space are fundamentally observer-dependent. In DQFT, this is built in at the level of the Hilbert space: different observers are associated with different sectors. In the ambient formalism, it appears through the non-existence of globally defined gauge-invariant local observables for gravitons. Together, they suggest that the correct framework for de Sitter quantum gravity will require a relational or holographic description in which information is intrinsically associated to causal patches rather than to a global spacetime [22, 23].

7 Conclusion and Future Directions

This paper has examined quantum field theory in de Sitter space through two complementary frameworks: Direct-Sum Quantum Field Theory, which reconstructs the quantum description from causal and symmetry principles, preserving unitarity; and the Ambient Space Formalism, which constructs fields as representations of $SO(1,4)$ in a higher-dimensional embedding, maintaining manifest covariance. The two frameworks address distinct aspects of the same underlying physics and, taken together, provide a richer picture than either affords alone.

Several themes emerge. The Gibbons–Hawking thermality (1) is not a signal of fundamental non-unitarity but of horizon-induced entanglement; the apparent mixed state perceived by a static observer reflects

causal restriction on an entangled pure state. Infrared divergences for massless fields can be traced, in the ambient language, to indecomposable representations of the de Sitter group—a structural property that makes clear which features are physical and which are coordinate artifacts. And the absence of a global vacuum or preferred time direction is not a defect but a feature: it points toward an inherently relational description of physics in accelerating universes.

A number of important questions remain open. The extension of both frameworks to interacting fields—where perturbation theory is complicated by secular growth—is an active area [24, 25]. The precise definition of gauge-invariant physical observables for gravitons in de Sitter space, and the relationship of de Sitter entropy to a microscopic theory, both await resolution. Finally, the question of whether exact de Sitter space can be a stable solution of a fundamental quantum gravity theory—as opposed to an approximate or asymptotic background—remains deeply open. Progress on these questions will likely require synthesis of the algebraic, information-theoretic, and holographic perspectives reviewed here.

De Sitter space is perhaps the simplest curved spacetime that forces every conceptual assumption of flat-space QFT to be re-examined. The challenges it presents are not obstacles but signposts pointing toward a more complete theory of quantum spacetime. For a broader treatment of the themes developed here, the reader is referred to the extended work [26].

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